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- The logo for the Smart India Hackathon (SIH) features a stylized brain. The left hemisphere is orange and contains a circuit board pattern with white lines and dots. The right hemisphere is green and contains white binary code (0s and 1s). Below the brain, the letters "SIH" are written in a bold, dark blue, sans-serif font. The entire logo is set against a white background with several dark blue diagonal lines radiating from the brain, suggesting a lightbulb or a network.

IDEA TITLE: NAUPLAN

Proposed Solution (Describe your Idea/Solution/Prototype)

NauPlan: a working prototype that moves SAIL from daily spot fixtures to short / medium term multi-voyage contracts for East Coast coal imports.

Detailed explanation of the proposed solution

Freight forecast: P10 / P50 / P90 ranges 1 to 26 weeks ahead, judged by walk-forward backtest

Vessel-port fit: Baltic standard ships vs draft, LOA, beam at 7 East Coast and 12 load ports

Contract strategy: optimiser picks spot, 3 / 6-month time charter or COA, and when to fix

Idle control: spare chartered days relet to the market; idle days tracked each month

Early warnings: sea state at all 7 ports, freight volatility, missing data flagged

How it addresses the problem

Replaces daily fixing with a contract plan tested on 200 freight scenarios

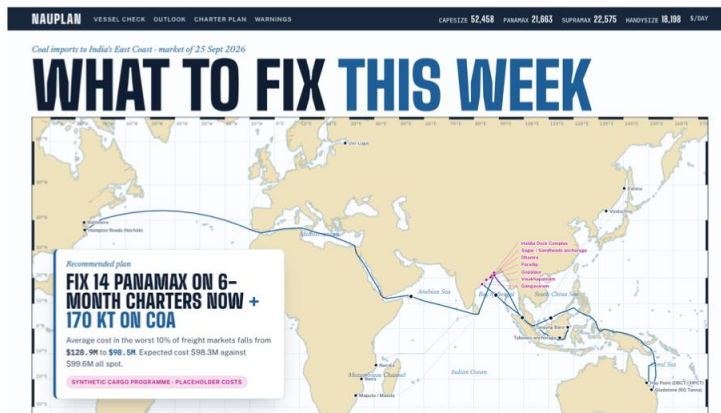
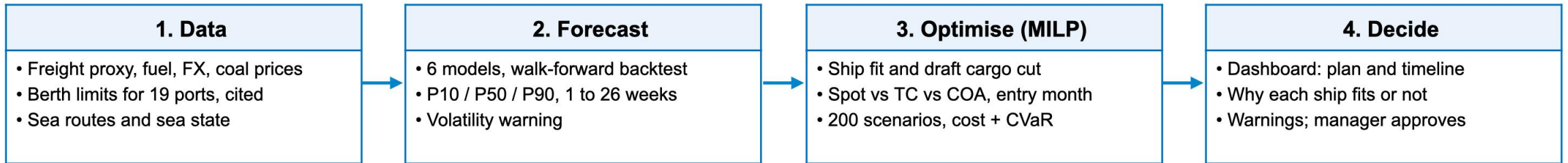
Shows why each ship can or cannot call, and the cargo cut at each draft limit

Innovation and uniqueness of the solution

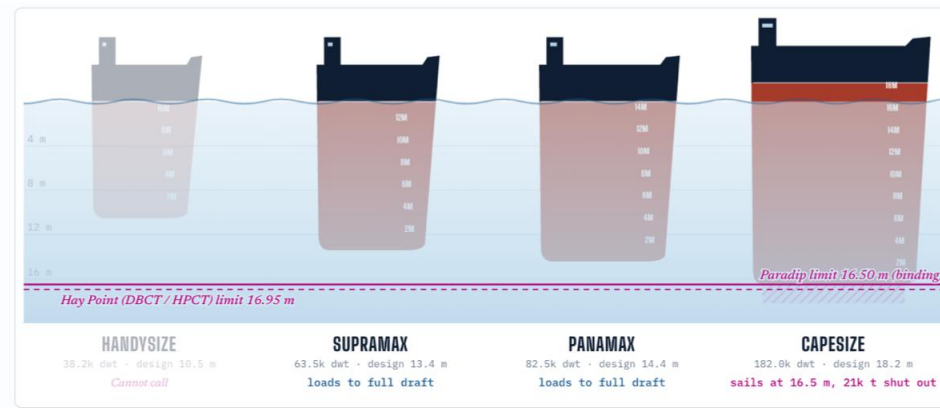
Plans on honest ranges and worst-case cost (CVaR), not on a single price guess

Every port limit cites its source; Paradip Capesize cargo within 1% of a real call

- **Technologies to be used (e.g. programming languages, frameworks, hardware)**
 - **Data & ML:** Python, pandas, statsmodels, LightGBM; FRED, World Bank, Open-Meteo, BDRY feeds
 - **Ports & routes:** cited berth limits for 19 ports; searoute (MARNET) with deep-draft and avoid-Suez routes
 - **Optimisation:** two-stage stochastic MILP in PuLP with the HiGHS solver; CVaR risk term
 - **App:** FastAPI backend, React + Vite + d3 dashboard; 37 automated tests; code on GitHub
- **Methodology and process for implementation (Flow Charts/Images/ working prototype)**



Prototype: recommended plan on a chart of voyage tracks



Draft gauge: which ship fits Hay Point to Paradip

Built and running

- One-command data refresh
- 19 ports, every limit cited
- Forecast backtest report
- Plan in seconds
- 37 automated tests
- Public code on GitHub

- **Analysis of the feasibility of the idea**
 - **Built:** data pipeline, cited limits for 19 ports, forecasting, optimiser, dashboard (37 tests)
 - **Checked:** Hay Point to Paradip Capesize, model 153,810 t vs 152,702 t actual call (6 Sep 2026)
 - **Tested:** forecasts backtested on 2021 to 2026 data; a 6-month plan solves in seconds
 - **Cost:** open-source stack on a laptop or one cloud VM; code public on GitHub
 - **Needs from SAIL:** import plan, fixture history, bunker and port costs (placeholders today)
- **Potential challenges and risks**
 - 1. Public data cannot predict freight direction (no model beat "no change" in testing)
 - 2. Per-class and period freight rates are licensed data
 - 3. Port limits change with tide and dredging; a few are not yet sourced
 - 4. Managers may not trust a black-box plan
- **Strategies for overcoming these challenges**
 - 1. Plan on ranges and scenarios; add the FFA forward curve with licensed data
 - 2. Baltic S8 route (Indonesia to East Coast India) or SAIL fixtures use the same pipeline
 - 3. Each limit stored with source and date; gaps shown on the dashboard
 - 4. Reasons for every ship choice; manager sets the risk weight; plan shown against all spot

- **Potential impact on the target audience**
 - **Chartering team:** a forward contract plan and ship choice with reasons, not daily market scanning
 - **Plant operations:** steadier coal arrivals, fewer stock-outs
 - **Port users:** ships sized to each port's draft, fewer waiting days
 - **Management:** expected and worst-case freight cost visible before committing
- **Benefits of the solution (social, economic, environmental, etc.)**
 - **Economic (example run*):** worst-10% programme cost \$128.9M all spot vs \$98.5M with the plan
 - **Operational:** spare chartered days relet, not idle; every decision auditable
 - **Environmental:** fewer ballast and waiting days means less fuel burnt

*Synthetic 3.88 Mt, 6-month programme with placeholder costs and 200 freight scenarios; real figures need SAIL data.

- **Details / Links of the reference and research work**
 - SIH 2026 problem statement SIH26006 (Ministry of Steel, SAIL)
sih.gov.in/sih2026PS
 - **Baltic Exchange: Guide to Market Benchmarks v8.7 (Aug 2026), standard vessels and TCE method**
balticexchange.com/.../ocean-bulk-guides-policies/GMB.pdf
 - **East Coast limits: Visakhapatnam Port Authority berth dimensions; Paradip Port Authority**
vpt.shipping.gov.in/admin_assets/uploads/1640237145_Berth_Details.pdf | paradiport.gov.in
 - **Adani Ports: Gangavaram BPTS (Oct 2024) and Dhamra BPTS (Oct 2025), berth parameters**
cdn.gac.com/prod/docs/INDIA-Gangavaram_New-BPTS-Oct-2024.pdf | adaniports.com (Dhamra BPTS)
 - **Load ports: DBCT Terminal Information Booklet (Hay Point); Port of Gladstone Handbook**
dbct.squarespace.com/s/Terminal-Information-Booklet-Procedure.pdf | gpcl.com.au
 - **Passages: Torres Pilots draft limits; Suez Canal Authority; Danish Maritime Authority**
torrespilots.com.au/pilot-info/draft-restrictions | suezcanal.gov.eg | soefartsstyrelsen.dk
 - **Data feeds: FRED (Brent, USD/INR); World Bank Pink Sheet; Open-Meteo ERA5 marine; BDRY**
fred.stlouisfed.org | worldbank.org/en/research/commodity-markets | open-meteo.com
 - **Methods: searoute (Eurostat MARNET); HiGHS solver; LightGBM; statsmodels**
github.com/genthalili/searoute-py | highs.dev | lightgbm.readthedocs.io
 - **Project repository, method notes and backtest report (Team Vector66)**
github.com/Tarkshya-26/nauplan